Statement of Accomplishments (SOA)

Sheng Di

TIME PERIOD: October 1, 2025 ~ September 30, 2026

1. Awards and honors

(1) 2026, Impact Argonne Award in the category of Enhancement of Argonne's Reputation in recognition of receiving a Best Paper at the SC25 11th International Workshop on Data Analysis and Reduction for Big Scientific Data.
(2) 2026, Impact Argonne Award in the category of Innovation in recognition of receiving the IEEE Technical Committee on Scalable Computing Award for Excellence (Middle Career Researcher).
(3) 2026, Impact Argonne Award in the category of Enhancement of Argonne's Reputation in recognition of your elevation to ACM Distinguished Members.
(4) 2026, Best paper award in the workshop ACM DC4AI2026 (in conjunction with ICPP2026).
(5) 2025, ACM Distinguished Member (announced in December, 2025).
(6) 2025, The IEEE Technical Committee on Scalable Computing (TCSC) award for Excellence (Middle Career Researcher) (announced in December, 2025).

2. Statement of accomplishments

The milestones and tasks established for 2025–2026 focused on six key areas: (1) submitting proposals to relevant DOE and NSF funding opportunities; (2) completing the planned research for ongoing projects; (3) maintaining and improving the associated software and libraries; (4) actively contributing to professional service and outreach; (5) strengthening existing collaborations while establishing new ones; and (6) actively mentoring students. All planned milestones and tasks were successfully completed.

Specifically, I submitted multiple proposals to NSF and DOE as planned, made substantial progress across all projects, significantly improved the performance, quality, and functionality of the associated software and libraries, and published numerous papers in leading conferences and journals. I also established and maintained a close communication with many application users from different projects to improve our compressors based on their demands.

As a summary, the major 2025-2026 achievement covers six aspects:

(1) **Submitting proposals to relevant DOE and NSF funding opportunities**: Contributed to **15** proposals—**10** DOE, four NSF, and one joint NSF–DOE proposal—with two DOE Genesis Mission proposals recommended for funding and three proposals pending.
(2) **Completing the planned research for ongoing projects**: Published or had accepted **39** peer-reviewed papers, including papers at SC, IPDPS, HPDC, ICS, VLDB, ICDE, ICLR, ICML, DSN, and IEEE VIS, in collaboration with researchers from 20+ institutions.
(3) **Maintaining and improving the associated software and libraries:** Made significant contributions to **20+** software libraries across **8** projects, including performance and quality improvements to cuSZp3/cuSZ, new evaluation functions in Z-checker/QCAT, QoZ 2.0 integration into SZ3, and the development of seven new tools. SZ-family software recorded 7,000+ unique downloads.
(4) **Maintaining active professional service and outreach**, including workshop leadership, approximately **10** program committees, two journal boards, a DOE review panel, nine presentations, an SC tutorial, and **12** support letters.
(5) **Strengthening existing collaborations while establishing new ones:** Established **10** new collaborations and maintained **34** existing ones, totaling **44** research collaborations across academia, national laboratories, and industry.
(6) **Actively mentoring students:** Mentored **38** students and early-career researchers, including four primary mentees and **34** co-mentees.

3. Research Products

3.1 Publications

From 10/01/2025 through 09/30/2026, I have **39 peer-reviewed papers published or accepted** in conferences and journals, including **6 IEEE/ACM SC 2026 papers, 6 IEEE IPDPS 2026 papers, 3 ACM HPDC 2026 papers, and 3 ACM ICS 2026 papers**, as well as 2 VLDB papers, 1 IEEE ICDE paper, 1 ICLR paper, 1 ICML paper, 1 IEEE DSN paper, 1 IEEE ICDCS paper, 2 IEEE Cluster papers, 1 IEEE/ACM CCGrid paper, 1 PASC paper, 1 IEEE VIS paper, 2 IEEE TPDS/TBD journal papers, and several workshop papers.

As for the research and software development, I am involved in **8** projects, including DOE ASCR ZF, NSF CSSI Framework FZ, NSF OAC Elements ROCCI, NSF CSSI OAC Elements SGCC, NSF CSSI OAC Elements Integration-NN, DOE Genesis Mission CASCADE, Genesis Mission Cerebras-code-generation, DOE PIONEER TIDES.

3.2 The published papers are listed as follows.

All papers listed here were published or accepted between 10/01/2025 and 09/30/2026 and have been archived in PANDA. Papers published in 2025 that were included in last year's performance appraisal have been excluded.

Conference papers:

(1) **APT# 206135:** Wenqi Jia, Zhewen Hu, Baixi Sun, Yafan Huang, Jiannan Tian, Boyuan Zhang, Daoce Wang, Sian Jin, Luanzheng Guo, **Sheng Di**, Yuede Ji, Miao Yin,"FLARE: A DataFlow-Aware and ScaLAble HardwaRE Architecture for Neural-Hybrid Scientific Lossy Compression." Paper accepted at the 32nd Asia and South Pacific Design Automation Conference (***ASP-DAC 2027***), null, January 25, 2027 - January 28, 2027. [acceptance rate: 29%]

(2) **APT# 201197:** Ning Yan, **Sheng Di**, Kai Zhao and Lipeng Wan. "Eureka: Enabling Fine-Grained Access and Range Queries on Compressed Scientific Data via Data-Index Co-Compression." Paper presented at the 52nd International Conference on Very Large Data Bases (***VLDB 2026***), August 31, 2026 - September 4, 2026.

(3) **APT# 205102:** Youyuan Liu, Longtao Zhang, Ruoyu Li, Bo Jiang, Taolue Yang, Kai Zhao, **Sheng Di**, et al. "Error-bounded Point Cloud Compression Using Truncated Octahedron Quantization." Paper presented at the 52nd International Conference on Very Large Data Bases (***VLDB 2026***), August 31, 2026 - September 4, 2026.

(4) **APT# 202346:** Ammar Ahmed, Azal Ahmad Khan, Ayann Ahmad, **Sheng Di**, Zirui Liu and Ali Anwar. "Retrieval-of-Thought: Efficient Reasoning via Reusing Thoughts." Paper presented at the 14th International Conference on Learning Representations (***ICLR 2026***), April 23, 2026 - April 27, 2026. [acceptance rate: 27%]

(5) **APT# 202056:** Bo Jiang, Taolue Yang, Youyuan Liu, Xubin He, **Sheng Di** and Sian Jin. "PackKV: Reducing KV Cache Memory Footprint through LLM-Aware Lossy Compression." Paper presented at the 40th IEEE International Parallel and Distributed Processing Symposium (***IEEE IPDPS2026***), May 25, 2026 - May 29, 2026. [acceptance rate: 23.27%]

(6) **APT# 202097:** Congrong Ren, Robert Underwood, **Sheng Di**, Emrecan Kutay, Zarija Lukic, Aylin Yener, Franck Cappello, et al. "FFCz: Fast Fourier Correction for Spectrum-Preserving Lossy Compression of Scientific Data." Paper presented at the 40th IEEE International Parallel and Distributed Processing Symposium (***IEEE IPDPS2026***), May 25, 2026 - May 29, 2026. [acceptance rate: 23.27%]

(7) **APT# 202100:** Shihui Song, Robert Underwood, **Sheng Di**, Peng Jiang and Franck Cappello. "Designing Domain-Specific Compilers for Lossy Compression: A Case Study on Wafer-Scale Engine." Paper presented at the 40th IEEE International Parallel and Distributed Processing Symposium (***IEEE IPDPS2026***), May 25, 2026 - May 29, 2026. [acceptance rate: 23.27%]

(8) **APT# 202337:** Pu Jiao, **Sheng Di**, Jiannan Tian, Mingze Xia, Xuan Wu, Yang Zhang, Xin Liang, et al. "Mitigating Artifacts in Pre-quantization Based Scientific Data Compressors with Quantization-aware Interpolation." Paper presented at the 40th IEEE International Parallel and Distributed Processing Symposium (***IEEE IPDPS2026***), May 25, 2026 - May 29, 2026. [acceptance rate: 23.27%]

(9) **APT# 202341:** Yuxiao Li, Mingze Xia, Xin Liang, Bei Wang, Robert Underwood, **Sheng Di**, Hemant Sharma, et al. "pMSz: A Distributed Parallel Algorithm for Correcting Extrema and Morse-Smale Segmentations in Lossy Compression." Paper presented at the 40th IEEE International Parallel and Distributed Processing Symposium (***IEEE IPDPS2026***), May 25, 2026 - May 29, 2026. [acceptance rate: 23.27%]

(10) **APT# 195781:** Shouwei Gao, Xu Zheng, Dongsheng Luo, **Sheng Di** and Wenqian Dong. "LUMOS: Democratizing SciML Workflows with L0-Regularized Learning for Unified Feature and Parameter Adaptation." Paper presented at the 40th IEEE International Parallel and Distributed Processing Symposium (***IEEE IPDPS2026***), May 25, 2026 - May 29, 2026. [acceptance rate: 23.27%]

(11) **APT# 202517:** Xuan Wu, **Sheng Di**, Tripti Agarwal, Kai Zhao, Xin Liang and Franck Cappello. "Enabling Homomorphic Analytical Operations on Compressed Scientific Data with Multi-stage Decompression." Paper presented at the 42nd IEEE International Conference on Data Engineering (***IEEE ICDE2026***), May 4, 2026 - May 8, 2026.

(12) **APT# 203123:** Shan Huang, Zhijing Ye, Jiamin Wang, Zhaorui Zhang, Jie Fu, Wendy Hui Wang, **Sheng Di**, et al. "In-Depth Exploration of Differentially Private Quantization for Privacy-Preserving and Communication-Efficient

Federated Learning." Paper presented at the 56th IEEE/IFIP International Conference on Dependable Systems and Networks (***IEEE DSN2026***), June 22, 2026 - June 25, 2026. [acceptance rate: 19.75%]

(13) **APT# 203779:** Shan Huang, Zhijing Ye, Zhaorui Zhang, **Sheng Di**, Jiamin Wang, Wendy Hui Wang and Xiaodong Yu. "Noise-Calibrated Lossy Compression for Privacy Leakage Reduction in Federated Learning." Paper presented at Federated Foundation Models at the Edge (***FFM@Edge***) workshop, in conjunction with the 46th IEEE International Conference on Distributed Computing Systems (***IEEE ICDCS2026***), June 22, 2026 - June 25, 2026.

(14) **APT# 203335:** Longtao Zhang, Ruoyu Li, Zhuoxun Yang, Robert Underwood, **Sheng Di**, Daoce Wang, Jinyang Liu, et al. "OPAL: On-demand Progressive Accelerated Scientific Lossy Compression." Paper presented at the 35th ACM International Symposium on High-Performance Parallel and Distributed Computing (***ACM HPDC2026***), July 13, 2026 - July 16, 2026. [acceptance rate: 14.2%]

(15) **APT# 203353:** Sujata Sinha, **Sheng Di**, Vishwas Rao, Robert Underwood, David Lenz, Zizhe Jian, Zhuoxun Yang, et al. "Bridging Information Theory and Practice for Scientific Lossy Compression." Paper presented at the 35th ACM International Symposium on High-Performance Parallel and Distributed Computing (***ACM HPDC2026***), July 13, 2026 - July 16, 2026. [acceptance rate: 14.2%]

(16) **APT# 203448:** Zhuoxun Yang, Ruoyu Li, Amit Subrahmanya, Vishwas Rao, **Sheng Di**, Robert Underwood, Longtao Zhang, et al. "TZ: Achieving High-Ratio Scientific Data Compression on GPUs with Global Data Decomposition." Paper presented at the 35th ACM International Symposium on High-Performance Parallel and Distributed Computing (***ACM HPDC2026***), July 13, 2026 - July 16, 2026. [acceptance rate: 14.2%]

(17) **APT# 202721:** Yafan Huang, **Sheng Di** and Guanpeng Li. "Not All Errors Are Equal: A Systematic Study of Error Propagation in Large Language Model Inference." Paper presented at the 2026 ACM International Conference on Supercomputing (***ACM ICS2026***), July 6, 2026 - July 9, 2026. [acceptance rate: 21%]

(18) **APT# 198292:** Ruoyu Li, Zhuoxun Yang, Jinyang Liu, Guanpeng Li, Yafan Huang, **Sheng Di**, Jiannan Tian, et al. "GPZ: GPU-Accelerated Lossy Compressor for Particle Data." Paper presented at the 2026 ACM International Conference on Supercomputing (***ACM ICS2026***), July 6, 2026 - July 9, 2026. [acceptance rate: 21%]

(19) **APT# 206057:** Yuke Li, Darren Ng, Arjun Kashyap, **Sheng Di**, Guanpeng Li, Xiaoyi Lu. "PACER: A Userspace Network Rate Controller in MPI with Adaptive Compression for Parallel Applications." Paper presented at the 2026 ACM International Conference on Supercomputing (***ACM ICS2026***), July 6, 2026 - July 9, 2026. [acceptance rate: 21%]

(20) **APT# 206202:** Adelina Zheng, Dong Dai, **Sheng Di**, Franck Cappello, Jinzhen Wang, "PRISM: Lightweight Surrogation for Lossy Compression Quality Prediction", Paper accepted at 11th International Parallel Data Systems Workshop (***PDSW2026***), 2026.

(21) **APT# 206200:** Milan Shah, **Sheng Di**, Michela Becchi, "Exploring Sparse Matrix Multiplication Kernels on the Cerebras CS-3." Paper accepted at 16th Workshop on Irregular Applications: Archictectures and Algorithms (***IA3***), 2026.

(22) **APT# 206204:** Yimeng Shan, Zhaorui Zhang, **Sheng Di**, Yu Liu, Xiaoyi Lu, Benben Liu, "SplitFT: An Edge-Cloud Collaborative Federated Split Learning System for LLM Fine-Tuning." Paper accepted at International Workshop on Edge-Cloud-HPC Operational Continuum (***ECHO2026***), 2026.

(23) **APT# 202539:** Bo Jiang, Taolue Yang, Youyuan Liu, Xubin He, Sheng Di and Sian Jin. "EmbdC: Error-Bounded Lossy Video Embedding Compression For On-Device LLM Inference." Paper presented at the 26th IEEE International Symposium on Cluster, Cloud, and Internet Computing (***IEEE/ACM CCGrid2026***), May 18, 2026 - May 21, 2026. [acceptance rate: 25.1%]

(24) **APT# 203750:** Viral Shah, Harikrishna Tummalapalli, Shivam Barwey, R. Balin, R. Balakrishnan, P. Fischer, **Sheng Di**, et al. "Solver-Integrated Lossy and Lossless Compression for Scalable Flow Simulations." Paper presented at ***PASC26***, June 29, 2026 - July 1, 2026.

(25) **APT# 206058:** Jin Lee, Zhonghao Chen, Xuhang He, Robert Underwood, Bogdan Nicolae, Franck Cappello, Xiaoyi Lu, **Sheng Di**, Zheng Zhang. "SPARe: Stacked Parallelism with Adaptive Reordering for Fault-Tolerant LLM Pretraining Systems with 100k+ GPUs." Paper presented at the International Conference on Machine Learning (***ICML2026***), 2026. [acceptance rate: 26.6%]

(26) **APT# 205044:** Jiefeng Zhou, **Sheng Di**, Yanfei Guo, Yuhao Guo, Lingqi Zhang, Kai Zhao, Rajeev Thakur, et al. "hZCCL2: Co-Designing Collective Communication with Vectorized Homomorphic Compression." Paper accepted at the 2026 International Conference for High Performance Computing, Networking, Storage, and Analysis (***IEEE/ACM SC2026***), November 15, 2026 - November 20, 2026. [acceptance rate: 19.2%]

(27) **APT# 205049:** Youyuan Liu, Bo Jiang, Taolue Yang, **Sheng Di**, Robert Underwood and Sian Jin. "TOPIQ: Statistical Error Propagation for Quantity-of-Interest Prediction under Lossy Compression." Paper accepted at the 2026 International Conference for High Performance Computing, Networking, Storage, and Analysis (***IEEE/ACM SC2026***), November 15, 2026 - November 20, 2026. [acceptance rate: 19.2%]

(28) **APT# 205064:** Changqing Li, **Sheng Di**, Kai Zhao and Wenqian Dong. "LLM Agents Meet Lossy Compression: Benchmark, Demystify and Optimize across HPC Architectures." Paper accepted at the 2026 International Conference for High Performance Computing, Networking, Storage, and Analysis (***IEEE/ACM SC2026***), November 15, 2026 - November 20, 2026. [acceptance rate: 19.2%]

(29) **APT# 205065:** Bohan Zhang, Yafan Huang, **Sheng Di** and Guanpeng Li. "From Error-Bounded to Fixed-Ratio: Efficient Sampling-Guided GPU Lossy Compression." Paper accepted at the 2026 International Conference for High Performance Computing, Networking, Storage, and Analysis (***IEEE/ACM SC2026***), November 15, 2026 - November 20, 2026. [acceptance rate: 19.2%]

(30) **APT# 205098:** Darren Ng, Abdullah Naveed, Duo Zhang, Guanpeng Li, **Sheng Di** and Xiaoyi Lu. "Z-SPDK: Accelerating Lossy Compressed I/O on NVMe-SSDs." Paper accepted at the 2026 International Conference for High Performance Computing, Networking, Storage, and Analysis (***IEEE/ACM SC2026***), November 15, 2026 - November 20, 2026. [acceptance rate: 19.2%]

(31) **APT# 205122:** Bo Jiang, Youyuan Liu, Taolue Yang, **Sheng Di** and Sian Jin. "3D Gaussian Splatting for Scientific Particle Data Compression and Rendering." Paper accepted at the 2026 International Conference for High Performance Computing, Networking, Storage, and Analysis (***IEEE/ACM SC2026***), November 15, 2026 - November 20, 2026. [acceptance rate: 19.2%]

(32) **APT# 205389:** Khondoker Mirazul Mumenin, Robert Underwood, Dong Dai, Jinzhen Wang, **Sheng Di**, Zarija Lukic and Franck Cappello. "DeepCQ: A Two-Stage Deep-Surrogate Framework for Lossy Compression Quality Prediction." Paper presented at the 2026 IEEE International Conference on Cluster Computing (***IEEE Cluster2026***), September 22, 2026 - September 25, 2026.

(33) **APT# 195995:** Ammar Ahmed, **Sheng Di**, Franck Cappello, Zirui Liu, Jingoo Han and Ali Anwar. "Systematic Evaluation of Optimization Techniques for Long-Context Language Models." Paper presented at the 2026 IEEE International Conference on Cluster Computing (***IEEE Cluster2026***), September 22, 2026 - September 25, 2026.

(34) **APT# 205568:** Guoxi Liu, Yuxiao Li, Congrong Ren, Robert Underwood, Xin Liang, Bei Wang, **Sheng Di**, et al. "FZ-VIS: A Visual Analytics Framework for Quantities-of-Interest-Aware Scientific Lossy Compression." Paper presented at IEEE Visualization Conference (***IEEE VIS2026***), November 9, 2026 - November 13, 2026. [acceptance rate: 24%]

(35) **APT# 201385:** Jiaqi Xu, Zhaorui Zhang, Gaolin Wei, **Sheng Di**, Benben Liu, Xiaodong Yu and Xiaoyi Lu. "DeepEBC: Compressing the Pre-Trained LLMs with Error-Bounded Lossy Compression." Paper presented at the 1st International Workshop on Big Data eXploration, Compression and Systems (***BDXCS2026***), January 26, 2026.

(36) **APT# 202722:** Changqing Li, Shouwei Gao, Kai Zhao, **Sheng Di** and Wenqian Dong. "Evaluating LLM Coding Agents on SZ-Family Lossy Compression Across Architectures." Paper presen**ted** at the 2nd Workshop on HPC for AI Foundation Models and LLMs for Science (***HPAI4S2026***), in conjunction with ***IEEE IPDPS2026***, May 26, 2026.

(37) **APT# 206055:** Tripti Agarwal, **Sheng Di**, Xin Liang, Zhaoyuan Su, Yuxiao Li, Hanqi Guo, and Franck Cappello. "TopoSZp: Lightweight Topology-Aware Error-controlled Compression for Scientific Data." Paper presented at the International Workshop on Data Compression for AI and Big Data Applications (***ACM DC4AI2026***) in conjunction with ***ACM ICPP2026***, 2026.

(38) **APT# 195874:** Zhonghao Chen, Weicong Chen, Duo Zhang, Kibaek Kim, Guanpeng Li, **Sheng Di**, Xiaoyi Lu, "FedDES: Discrete Event Performance Simulation for Large-Scale Federated Learning Systems", in ACM/IEEE Symposium on Edge Computing (***SEC2025***), 2025.

Journal Articles:

(1) **APT# 204948:** Arham Khan, **Sheng Di**, Kai Zhao, Jinyang Liu, Kyle Chard, Ian T. Foster and Franck Cappello. "SLyCE: An Efficient Surrogate-based Lossy Compression Quality Estimation Framework.", IEEE Transactions on Big Data (***IEEE TBD***), vol. 12, no. 5, pp. 2002-2017, Oct. 2026.

Poster presentation:

(1) **APT# 202723**: Xuan Wu, **Sheng Di**, Tripti Agarwal, Kai Zhao, Xin Liang and Franck Cappello. "Enabling Homomorphic Analytical Operations on Compressed Scientific Data with Multi-stage Decompression." Abstract of presentation at the 42nd IEEE International Conference on Data Engineering (***IEEE ICDE2026***), Montreal, May 4, 2026 - May 8, 2026.

(2) **APT# 202880: Sheng Di**, Vishwas Rao, David Lenz, Amit Subrahmanya, Robert Underwood, Zhuoxun Yang, Zizhe Jian. "ZF Thrust 3: Novel compressor designs and evaluation." Abstract of presentation at the FZ ZF Joint workshop at USF, Tampa, Florida, March 17, 2026 - March 18, 2026.

(3) **APT# 206006:** Cappello, Franck, **Sheng Di**, Robert Underwood, Fengguang Song, Hanqi Guo and Kai Zhao. "FZ: A fine-tunable cyberinfrastructure framework to streamline specialized lossy compression development." Poster presented at the 2026 NSF CSSI / Cyber Training / SCIPE PI Meeting (***NSF PI Meeting***), Rockville, Maryland , September 13, 2026-September 15, 2026.

Slides/Oral Presentation:

(1) **APT# 201754:** Shihui Song, Robert Underwood, Sheng Di, Peng Jiang and Franck Cappello. "Leveraging Wafer Scale Engine for Lossy Compression." paper presentation at the International Conference on High Performance Computing in the Asia-Pacific Region, Osaka, Japan, January 26, 2026 - January 29, 2026.

(2) **APT# 202627:** Sheng Di, Yafan Huang, Shihui Song, Robert Underwood, Peng Jiang and Franck Cappello. "Error-bounded Lossy Compression for Accelerators." Institutional Talk at Tohoku University, March 9, 2026.
(3) **APT# 202672:** Sheng Di. "Recent Advances in Error-Bounded Lossy Compression with SZ for Scientific and AI Workloads." Institutional Talk at Forschungszentrum Julich, May 21, 2026.
(4) **APT# 202802:** Sheng Di, Changqing Li, Shouwei Gao, Kai Zhao and Wenqian Dong. "Evaluating LLM Coding Agents on SZ-Family Lossy Compression Across Architectures." paper presentation at the 40th IEEE IPDPS 2026, New Orleans, LA, May 25, 2026 - May 29, 2026.
(5) **APT# 203692:** Sheng Di. "Towards Scalable AI: Leveraging Lossy Compression Across the Stack." Slide presentation at the 18th JLESC Workshop, May 19, 2026 - May 21, 2026.
(6) **APT# 204883:** Sheng Di. "Towards Error-bounded Lossy Compression for Science and AI." invited talk at the tutorial "When the Error-Bounded Lossy Compression Meets Large-Scale AI Model Training and Fine-Tuning in Federated Environments", in conjunction with ACM HPDC 2026, Cleveland, OH, July 13, 2026 - July 16, 2026.
(7) **APT# 204885:** Sheng Di. "Scalable Dynamic Scientific Data Reduction." Invited talk at the Early Career Panel of 2026 DOE Computer Science Principal Investigators (PI) Meeting, Washington, DC, July 21, 2026 - July 22, 2026.
(8) **APT# 204886:** Sheng Di. "Compression-aware resilience for federated and AI-driven workloads." Invited talk at the breakout session (Theme 4: Trustworthy AI, Software Integrity, Resilience, and Strategic Community Cultivation) of the 2026 DOE Computer Science PI Meeting, July 21, 2026 - July 22, 2026.
(9) **APT# 205101:** Sheng Di. "Error-Bounded Lossy Compression with SZ for Scientific and AI Workloads." Invited talk at the International Workshop on Data Compression for AI and Big Data Applications (DC4AI), in conjunction with 55th International Conference on Parallel Processing (ACM ICPP2026), September 28, 2026 - October 1, 2026.
(10) **APT# 206972:** Robert Underwood, Sheng Di, Jiannan Tian, Yafan Huang, Hanqi Guo, Yuxiao Li, Congrong Ren and et al."Situational Awareness: Advances in Compression for Light Sources." Slide-presentation at the Illumine All Hands Meeting, Melno Park, CA USA, August 18, 2026 - August 19, 2026.
(11) **APT# 206973:** Robert Underwood, Jiannan Tian, Kevin Dougherty, Skyler Ruiter, Sujata Sinha, Lingja Liu, Vishwas Rao, David Lentz, Zizhe Jian, Zhuoxun Yang, Kai Zhao, Sheng Di, Franck Cappello, "Encode in Place (EIP), Information Theory, and Compression for Globus." Slide presentation at the Illumine All Hands Meeting, Melno Park, CA USA, August 18, 2026 - August 19, 2026.
(12) **APT# 206976:** Underwood, Robert, Jon Calhoun, Sheng Di and Franck Cappello. "Understanding The Effectiveness of Lossy Compression in Machine Learning Training Sets." Slide presentation at the 2026 SIAM Conference on Uncertainty Quantification, Minneapolis, Minnesota United States, March 22, 2026 - March 25, 2026.

3.3 Projects and Technical Leads

(1) **PI (Argonne institute PI)** of DOE ECRP SDR project: Scalable Dynamic Scientific Data Reduction, 2021~2026.
(2) **co-PI (Argonne institute PI)** of DOE Genesis Mission project Phase I: An Agentic LLM Workflow for Code Generation and Optimization on Emerging AI Accelerators Phase I Proposal, 2026-2027.
(3) **co-PI (Argonne institute PI)** of DOE Genesis Mission project Phase I: CASCADE: Composable Assurance for AI-enabled Scientific Computing with Agentic Error Governance, 2026-2027.
(4) **co-PI (Argonne institute PI)** of DOE ASCR distributed resilience project: Scalable and Resilient Modeling for Federated-Learning-Based Complex Workflows (PI: Xiaoyi Lu), 2023~2028.
(5) **co-PI** of NSF Collaborative Research: Frameworks: FZ: A fine-tunable cyberinfrastructure framework to streamline specialized lossy compression development (PI: Franck Cappello), 2023-2028.
(6) **co-PI (Uchicago institute PI)** of NSF Collaborative Research: Elements: SGCC: An Efficient GPU-oriented Data Reduction Cyberinfrastructure for Scientific Data Analysis, 2025-2028.
(7) **Senior personnel** of DOE ASCR ZF: A novel framework to design trustworthy lossy compressors for scientific data approaching lossy compressibility limits (PI: Franck Cappello), 2024-2027.
(8) **Senior personnel** of NSF Collaborative Research: Elements: A Cyberinfrastructure for Seamlessly Integrating Lossy Compression into Neural Networks for Scientific Applications (PI: Michela Becchi, Franck Cappello), 2025-2028.
(9) **Senior personnel** of DOE PIONEER TIDES project: TIDES4AI: Transformative Integration of Scientific Data for AI, 2025-2030.

3.4 Other projects I participated in

(1) **Collaborator and Student coadviser**: DOE ILLUMINE project - Advanced Computing Technologies for Integrated and Optimized Neutron scattering and light source experiments (PI: Jana Thayer)
(2) **Collaborator and Student coadviser**: DOE SWARM project - Exploring the Power of Distributed Intelligence for Resilient Scientific Workflows (PI: Ewa Deelman)

4. Software

As for the software development, I participated in the development and maintenance of many existing libraries/projects, in the regard of lossy compression, data assessment, communication (including SZ1-3, FZ, SZx/cuSZx, SZ3, cuSZ, cuSZp2/3, SZp, Z-checker, cuZ-checker, QCAT, FXRZ, C-Coll, gZCCL, hZCCL, FAZ, FZ-GPU, SECRE, XTIMATE,

IPComp, Ocelet, COMPSO, ghZCCL, QPET, etc.). I also contributed to the design and development of multiple new libraries/toolkits (including SLyCE, ZCCL2, UMC, LCP, TempGraph, TurboFFT, etc.). Typical examples are summarized below (yet not limited). The SZ-family products have 7000+ unique downloads according to github record.

The software products under optimization and maintenance in 2025-2026:

(1) **SZ2.** Error-bounded lossy compressor. https://github.com/szcompressor/SZ2
(2) **SZ3**. Composable Error-bounded Lossy Compression Framework. https://szcompressor/SZ3
(3) cuSZp3. I lead our team to develop an ultra-fast lossy compressor - namely cuSZp3 (GPU version), which improved the compression and decompression speed significantly over the cuSZ, with outstanding compression ratios and significantly mitigated artifact issues compared with the cuSZx. https://github.com/szcompressor/cuSZp
(4) **cuSZ.** I participated in the maintenance and improvement of the cuDA version for SZ (including improving the Huffman encoding/decoding performance, GPU version of LZ77). https://github.com/szcompressor/cuSZ
(5) **Libpressio-family.** I co-developed the Libpressio-family products including Libpressio, LibpressioOpt, Libpressio-Predict, with my students and a postdoc. Libpressio (https://github.com/robertu94/libpressio) is an abstract software that can be used to call different compressors (supporting over 30 now) in a uniform interface; LibpressioOpt (https://github.com/robertu94/libpressio_opt) automatically searches for the best parameter settings for the user-customized metric.
(6) **Z-checker/Qcat.** I kept maintaining the Z-checker and its lightweight version Qcat, by fixing a series of potential bugs and adding more important functions such as calculating various derivatives (Sobolev norms, etc.). https://github.com/CODARcode/z-checker-installer ; https://github.com/CODARcode/z-checker ; https://github.com/szcompressor/QCAT
(7) **SZ/cuSZx.** I kept improving cuSZx – the GPU version of ultra-fast error-bounded lossy compressor (SZx) by enabling it to support accurate estimation of compression ratio and other compression quality metrics. https://github.com/szcompressor/SZx
(8) **FXRZ.** I kept leading the development of the FXRZ framework (Fixed-ratio compression). https://github.com/hasanur-rahman/FXRZ
(9) **QoZ.** I kept leading the development of Quality-oriented error-bounded compression (QoZ2.0). We integrated QoZ2.0 into the SZ3 framework. https://github.com/JLiu-1/HPEZ
(10) **SDRBench.** Scientific Data Reduction Benchmark. https://sdrbench.github.io/
(11) **ROCCI**. I kept leading and improving the ROCCI project/library. https://roccilab.github.io/rocci/
(12) **UMC.** I participated in the design and development of UMC – error-bounded lossy compressor optimized for unstructured datasets. The paper received the best paper award in IPDPS 2025. https://github.com/xuanwu02/umc
(13) **QPET.** A Versatile and Portable Quantity-of-Interest-preservation Framework for Error-Bounded Lossy Compression. https://github.com/JLiu-1/QPET-Artifact
(14) **IPComp.** Interpolation Based Progressive Lossy Compression for Scientific Applications. https://github.com/hpdslab/IPComp

The new software products developed in 2025-2026:

(1) **SlyCE.** As the project PI, I led the development of SLyCE: A Surrogate-based Lossy Compression Quality Estimation Framework. The paper under review in TPDS. The code will be released with the paper acceptance later. https://roccilab.github.io/rocci/
(2) **hZCCL2**. I led the development of hZCCL2: Co-Designing Collective Communication with Vectorized Homomorphic Compression (accepted by SC2026). https://zccl.org/
(3) **TZ**. I participated in the design and development of TZ: Achieving High-Ratio Scientific Data Compression on GPUs with Global Data Decomposition (accepted by HPDC2026). https://github.com/hpdslab/tensorCodec
(4) **FFCz**. I participated in the design and development of FFCz: Fast Fourier Correction for Spectrum-Preserving Lossy Compression (accepted by IPDPS2026). https://github.com/streamlineLab/FFCz
(5) **OPAL**. On-demand Progressive Accelerated Scientific Lossy Compression. I participated in the design and development of OPAL: On-demand Progressive Accelerated Scientific Lossy Compression (accepted by HPDC2026). https://github.com/LangdaoZhang/libparz and https://github.com/LangdaoZhang/parSZ
(6) **GPZ**. GPU-Accelerated Lossy Compressor for Particle Data. https://github.com/hpdslab/GPZ
(7) **pMSz**, a parallel topology-preserving corrector (Morse-Smale segmentations) for lossy compression, https://github.com/streamlineLab/pMSz

5. Professional Activities and Recognition

I served as the program chair for 1 workshop:

(1) Program Co-chair: 1st International Workshop on Big Data eXploration, Compression and Systems (BDXCS) in conjunction with SCA/HPCAsia2026.

I organized and served as the steering committee chair for 2 workshops:

(1) The 6th International Workshop on Big Data Reduction (IWBDR). https://mercury-hpg.github.io/IWBDR-6/
(2) The 1st International Workshop on Edge-Cloud-HPC Operational Continuum (ECHO). https://echo-workshop.github.io/#organization

I served in the program committee board in 10 conferences/workshops, including:

(1) **Computer Science Reviewing Board Member**: 13th Heidelberg Laureate Forum (HLF), 2026.
(2) **Program Committee Member:** IEEE International Conference on Cluster Computing (IEEE CLUSTER 2026).
(3) **Program Committee Member:** International Conference on Supercomputing (ACM ICS 2026).
(4) **Program Committee Member:** International Workshop on Edge-Cloud-HPC Operational Continuum (IEEE/ACM ECHO2026) in conjunction with IEEE/ACM The International Conference for High Performance computing, Networking, Storage and Analysis (IEEE/ACM SC2026).
(5) **Program Committee Member:** AAAI Conference on Artificial Intelligence (AAAI 2027)
(6) **Program Committee Member):** IEEE/ACM The International Conference for High Performance computing, Networking, Storage and Analysis (IEEE/ACM SC2026), 2026, poster session.
(7) **Program Committee Member:** IEEE International Symposium on Workload Characterization (IISWC 2026).
(8) **Program Committee Member:** 26th IEEE International Symposium on Cluster, Cloud, and Internet Computing (CCGrid 2026).
(9) **Program Committee Member:** The 6th InternationalWorkshop on BigData Reduction (IWBDR 2026).

I served in the proposal review board:

(1) Invited panel member to review a couple of proposals submitted for DOE EPSCoR funding opportunity

I am in the editorial/review board in two journals:

(1) Editorial Board Member, Frontiers in High Performance Computing Journal (FHPCJ).
(2) Review Board Member, Transactions on Parallel and Distributed Systems (TPDS).

I provided multiple support/reference letters for the students/colleagues to apply for different programs:

(1) Jiannan Tian — Collaboration letter for an NSF proposal.
(2) Xiaoyi Lu — Nomination letter for the IEEE TCSC Middle Career Award.
(3) Di Zhang — Reference letter for a postdoctoral position.
(4) Miao Yin — Collaboration letter for a PESOSE proposal.
(5) Xuan Wu — Recommendation letter for the Rising Star Award in Data Science. Xuan received the award, which will be presented during the SC26 awards ceremony.
(6) Lipeng Wan — Collaboration letter for an NSF CAREER proposal.
(7) Ali Anwar — Collaboration letter for an NSF CAREER proposal.
(8) Hongyuan Liu — Collaboration letter for an NSF CAREER proposal.
(9) Kevin Dougherty — Reference letter for a predoctoral position at Argonne National Laboratory.
(10) Yafan Huang — Reference letter for faculty position applications. Yafan received an Assistant Professor offer from the University of Maryland.
(11) Shihui Song — Reference letter for faculty position applications. Shihui received an Assistant Professor offer from Wayne State University.
(12) Pu Jiao — Reference letter for faculty position applications. Pu received an Assistant Professor offer from the University of Texas Rio Grande Valley

I served as an external invited reviewer for conferences/journals/NSF, including:

(1) Reviewer: IEEE Transactions on Parallel and Distributed Computing (IEEE TPDS), 2026.
(2) Reviewer: IEEE Transactions on Computers (IEEE TC), 2026.
(3) Reviewer: International Journal of Future Generation Computer Systems (FGCS), 2026.
(4) Reviewer: ACM Transactions on Internet of Things (ACM TIoT), 2026.
(5) Reviewer: Journal of Parallel Computing (PARCO), 2026.
(6) Reviewer: IEEE Transactions on Knowledge and Data Engineering (IEEE TKDE), 2026.
(7) Reviewer: Journal of Advances in Modeling Earth Systems, 2026.
(8) Reviewer: Eurographics Conference on Visualization (Eurovis), 2026.

I served in CELS interview committee for the following employees:

(1) Alvin Gonzales (Postdoc -> RD2 interview)

I served in Ph.D defense board for the following students:

(1) Tripti Agarwal (University of Utah)
(2) Alex Fallin (Texas State University)

6. Proposals

I actively participated in the writing of 15 proposals. Two of them have been recommended for funding. Three of them are still under the pending state.

(1) **Argonne Institute PI:** DOE Genesis Mission project Phase I (partner-led): An Agentic LLM Workflow for Code Generation and Optimization on Emerging AI Accelerators Phase I Proposal, 2026-2027. (**Encouraged**)
(2) **Argonne Institute PI:** DOE Genesis Mission project Phase I (partner-led): CASCADE: Composable Assurance for AI-enabled Scientific Computing with Agentic Error Governance, 2026-2027. (**Encouraged**)
(3) **overall lead PI:** NSF Cybertraining: **(Pending)** We received a funding recommendation in June but was placed in pending status in August due to limited funding availability
(4) **Argonne Institute PI:** NSF-DOE CS2 proposal: (Argonne institute PI): (**pending**)
(5) **Argonne Institute PI:** NSF Future CoRe: (**pending**)
(6) **UChicago institute PI:** NSF Collaborative Research: SHF: HZ: a generic error-controlled compression framework with support for diverse operations on compressed data (PI: Kai Zhao, Sheng Di and Xin Liang) $900,000 for 3 years. (**Discouraged**)
(7) **UChicago institute PI:** NSF CSSI Elements OAC proposal: co-PI (Uchicago institute PI): Collaborative Research: Elements: XComMPI: An Extensible Cyberinfrastructure for Compression-Accelerated MPI in Scientific Applications (**Discouraged**)
(8) **Argonne institute PI:** DOE Genesis Mission project Phase I: Metacognitive Reuse for Trustworthy Mathematical and Symbolic Reasoning in Scientific Foundation Models (**Discouraged**)
(9) **Argonne institute PI:** DOE Genesis Mission project Phase I: VERITAS: Verified and Reliable AI for Trustworthy Scientific Software in HPC Environments (**Discouraged**)
(10) **Argonne institute PI:** DOE Genesis Mission project Phase I: TwinNet: AI-Driven Digital Twin for Immersive Autonomous Operations in DOE Research Networks (**Discouraged**)
(11) **Argonne institute PI:** DOE Genesis Mission project Phase I: Composable Atomistic Foundation Models with HPC-Aware Orchestration(**Discouraged**)
(12) **Argonne institute PI:** DOE Genesis Mission project Phase I: Composable Scientific Reasoning Models and Workflows for Closed-Loop Discovery (**Discouraged**)
(13) **Argonne institute PI:** DOE Genesis Mission project Phase I: Toward Fault-Aware Trustworthy AI-Driven Scientific Workflows (**Discouraged**)
(14) **Argonne institute PI:** DOE Genesis Mission project Phase I: Framework for Neuro-Symbolic Agents in High-Performance Computing Code Development (**Discouraged**)
(15) **co-PI:** DOE Genesis Mission project Phase I: CASCADE: Cross-Stack Agentic Software and Hardware Co-Design for HPC Applications with Near-Memory Data Execution (**Discouraged**)

I am in charge of the compute resource allocation renewal and project report for our team, including SDR project and FZ project

(1) LCRC Bebop: 2325 core-hours
(2) LCRC Swing: 500 node-hours
(3) LCRC Improv: 2772 node-hours
(4) ALCF Polaris: ~3000 node-hours per quarter
(5) ALCF Sophia: 150 node-hours per quarter
(6) ALCF Aurora: 1000 node-hours per quarter
(7) NERSC: 500 node-hours per quarter
(8) ACCESS-CI: 800000 access credits (for NSF project)

7. Student Mentoring

In this FY year, I majorly mentored 4 students, and co-mentored 34 other students under multiple collaborative projects.

My majorly mentored students are listed as follows:
For these students, I had separate weekly meetings with each of them during the working period, and also significantly got involved
in the research design, implementation, performance evaluation, and paper writing.

Student	Period	Main Research and Topic	Related Projects
Md Hasanur Rahman	10/01/2024 ~ present	Develop/optimize compression quality estimation engine/algorithm	DOE ECRP
Tripti Agarwal	10/01/2024 ~ 06/30/2026	Homomorphic compression and topology-preservation in lossy compression	DOE ASCR auto data reduction
Zizhe Jian	10/01/2024 ~ present	Optimization of lossy compression in ZF project: including the exploration more efficient compression algorithms and optimization for domain applications	DOE ECRP and ZF
Yue Zhu	05/15/2026 ~ present	Characterizing and improving federated learning for RS-APPFL project	DOE RS-APPFL

33 students co-mentored with other researchers in this FY year:
For these students, I attended weekly or bi-weekly meetings hosted by me or other mentors during the working period, and also participated in the design, discussion, resolving technical issues, coding, and paper writing.

(1) Arham Khan (10/01/2025 ~ 06/30/2026): Develops the SLyCE framework, which extends the SECRE framework to support multiple lossy compression quality metrics under the surrogate model. (Arham participates in multiple projects in this period, including the ROCCI project that I served as the PI)
(2) Yafan Huang (10/01/2025 ~ 06/30/2026): Improve compression performance for GPU and quality for Illumine project, with Robert Underwood and Franck Cappello.
(3) Milan Shah (10/01/2025~ 2026/06/30): Collaboration on compression for Qtensor and performance improvement, with Xiaodong Yu, Michela Becchi (NCSU), Franck Cappello, Yuri Alexeev, etc.
(4) Ning Yan (10/01/2025 ~ 09/30/2026): Compression + Query in ADIOS2, with Lipeng Wan (College of Arts & Sciences).
(5) Jiao Pu (10/01/2024 ~ 06/30/2026): QoI-preserved compression; detecting, Mitigating and removing artifacts for compression, with Xin Liang (U. Kentucky) and Franck Cappello.
(6) Khondoker Mumenin (10/01/2025 ~ 09/30/2026): Surrogate for lossy compression result, with Dong Dai and Jinzhen Wang.
(7) Zhonghao Chen (10/01/2025 ~ 09/30/2026): Simulation of Federated Learning, with Xiaoyi Lu.
(8) Shixun Wu (10/01/2025 ~ 06/30/2026): Swarm intelligence in the SWARM project, with Franck Cappello and Christina.
(9) Shihui Song (10/01/2025 ~ 09/30/2026): Development of compressor for Serebras (CereSZ-II) and compression compiler for FZ project, with Peng Jiang, Robert Underwood and Franck Cappello.
(10) Weijin Liu (10/01/2025 ~ 09/30/2026): Characterization of LLM performance, with Xiaodong Yu (SIT).
(11) Ammar Ahmed (10/01/2025 ~ 09/30/2026): Compression for LLM, with Ali Anwar (UMN) and Franck Cappello.
(12) Congrong Ren (10/01/2025 ~ 09/30/2026): Optimization for particle dataset and spectrum-preservation, with Hanqi Guo (OSU), Robert Underwood and Franck Cappello.
(13) Longtao Zhang (10/01/2025 ~ 09/30/2026): Optimization for particle dataset, with Kai Zhao (FSU).
(14) Ruoyu Li (10/01/2025 ~ 09/30/2026): Optimization GPU version of particle data compression and progressive compression, with Kai Zhao (FSU).
(15) Shan Huang (10/01/2025 ~ 09/30/2026): Security in Federated Learning, with Xiaodong Yu (SIT).
(16) Xuan Wu (10/01/2025 ~ 09/30/2026): Homomorphic compression with supported stencil operations and MPI+compression, with Xin Liang (UKY).
(17) Bohan Zhang (10/01/2025 ~ 09/30/2026): Lossy checkpointing and fixed-ratio compression, with Guanpeng Li.
(18) Abdullah Naveed (10/01/2025 ~ 09/30/2026): Developing highly efficient symptom-based soft error detector for HPC applications, with Guanpeng Li.
(19) Giovanni Rivera (10/01/2025 ~ 09/30/2026): Resilience research for AI/LLM, with Zizhong Chen and Franck Cappello.
(20) Ruiheng Chen (10/01/2025 ~ 09/30/2026): Enabling PnetCDF to support lossy compression, with Fengguang Song and Weikeng Liao.
(21) Ziwei Qiu(10/01/2025 ~ 09/30/2025): Characterize the various lossy compressors for correlation-based compression metrics and improve compression quality estimation method, with Hanqi Guo and Robert Underwood.

(22) Youyuan Liu (10/01/2025 ~ 09/30/2026): Explore feature deduplication for LLM, with Sian Jin and Robert Underwood.
(23) Bo Jiang (10/01/2025 ~ 09/30/2026): Compression for KV Cache in LLM, with Sian Jin..
(24) Hojin Lee (10/01/2025 ~ 06/30/2026): Resilience for LLM, with Robert Underwood, Bogdan N., and Franck Cappello.
(25) Sujata Sinha (10/01/2025 ~ 09/30/2026): Explore lossy compressibility in theory and improve compression quality, with Vishwas, Robert Underwood and Franck Cappello.
(26) Yuke Li (10/01/2025 ~ 09/30/2026): User-level network control for MPI operations, with Xiaoyi Lu.
(27) Jiefeng Zhou (10/01/2025 ~ 09/30/2026): Enabling Compression Capability in Modern MPI Libraries, with Jiajun Huang and Yanfei Guo
(28) Kevin Dougherty (05/01/2025 ~ 09/30/2026): Enabling HDF5 to support GPU compressors; improving scalability for quantum circuit simulation with compression, with Robert Underwood.
(29) Yiwenbo Pu (08/01/2026 ~ 09/30/2026): Improving lossy compression quality and performance for GPU by R-ANS arithmetic algorithm, with Jiajun Huang.
(30) Yuhao Guo (06/01/2026 ~ 09/30/2026): Improving stencil computation performance and agentic system for benchmark environment, with Jiajun Huang and Lingqi Zhang (RIKEN)
(31) Changqing Li (01/01/2026 ~ 09/30/2026): Developing an agentic system generating compression code for AI accelerators (GPU, Cerebras), with Wenqian Dong and Kai Zhao.
(32) Tian Qiu (04/01/2026 ~ 09/30/2026): Neural Compression for Scientific Particle Data with Haewon Jeong (UCSB).
(33) Wenqi Jia (05/01//2026 ~ 09/30/2026): Exploiting Cross-trajectory Recurrence to Accelerate Information-seeking RL Rollouts over Shared Corpora, with Miao Yin.
(34) Ben Mechels (07/30/2026 ~ 09/30/2026): Parallel Transform-based Codec for Efficient Asymmetric Signal Compression, with Kyle Chard.

8. Teaching/Outreach

(1) I am one of the organizers of the tutorial 'lossy compression for scientific data', which was held in SC 2025 and to be held in SC 2026.
(2) I collaborated with a Virginia Tech news writer to develop a feature story on the FZ project. The news has been posted here: https://news.vt.edu/articles/2025/08/compression-ratio-smart-glasses-satellites-virginia-tech-argonne.html
(3) I attended the JLESC workshop in Germany to build and sustain international collaborations.
(4) I attended the NSF CSSI PI meeting to present two CSSI OAC projects on which I serve as co-PI.
(5) At the DOE PI meeting, I presented a poster on my ECRP project and served as an invited panelist, sharing my experience as a mid-career researcher.
(6) I attended the DOE–MEXT meeting to report on and discuss the collaboration between Argonne and RIKEN.

9. Collaboration

New collaborations were established in 2025-2026, including:

(1) Ali Anwar (University of Minnesota): Optimization and efficient reasoning for long-context LLMs.
(2) Bei Wang (University of Utah): Topology- and QoI-aware scientific lossy compression and visual analytics.
(3) Vishwas Rao (Oak Ridge National Laboratory / University of Tennessee): Information-theoretic foundations and modeling for scientific lossy compression.
(4) Wenqian Dong (North Carolina State University): LLM agents for scientific lossy compression and SciML workflow optimization.
(5) Dongsheng Luo (Florida International University): Machine learning and SciML workflow optimization.
(6) Miao Yin (University of Texas at Arlington): Hardware architectures for neural-hybrid scientific lossy compression.
(7) Lipeng Wan (Georgia State University): Fine-grained access and range queries over compressed scientific data.
(8) Lingjia Liu (Virginia Tech): Information-theoretic foundations for scientific lossy compression.
(9) Viral Shah (JuliaHub / Julia Computing): Solver-integrated lossy and lossless compression for scalable scientific simulations.
(10) Xubin He (Temple University): LLM-aware lossy compression for KV cache and video embeddings.

Multiple collaborations were maintained or strengthened in 2025-2026, including:

(1) Zheng Zhang (UCSB): Co-mentoring Hojin Lee on the resilience research for LLM.
(2) Weikeng Liao (Northwestern U.): integrating compression into PnetCDF.
(3) Jinzhen Wang (UNC Charlotte): Develop efficient quality estimation method for lossy compressors.

(4) Viral Sudip Shah (UIUC) & Ramesh Balakrishnan (CPS, Argonne): Optimize lossy compression quality for Combustion datasets (Flamespeed).
(5) Yan Feng (EVS, Argonne): Optimize compression for Climate datasets.
(6) kaz Yoshii (MCS, Argonne): Optimize compression for hardware/chips.
(7) Haomin Yuan (NSE, Argonne): Optimize compression for Net5000/CFD with unstructured datasets.
(8) OceanBox team (OceanBox company): Optimize compression for Climate application with unstructured datasets.
(9) Pinaki Pal (TAPS, Argonne): Optimize compression for CFD applications.
(10) Robert Jackson (EVS, Argonne): Lidar data compression with preservation of spectra and moments.
(11) Ali Anwar (UMN): Compression for LLM.
(12) Kai Zhao (FSU): High-speed Particle data compression and Progressive Compression
(13) Hanqi Guo (OSU): High-ratio Particle data compression.
(14) Xiaodong Yu (SIT): Characterization of LLM performance, and privacy preservation in federated learning.
(15) Wenqian Dong (Oregon State U.): Improving LLM performance on various architectures including AI testbeds.
(16) Amit Bashyal (ANL): Compression for DUNE dataset.
(17) Thomas Grützmacher(KIT): using compression in linear algebra library and optimizing performance.
(18) Ian Foster/Kyle Chard (U. Chicago): Deploying lossy compressor on Globus and improving overall performance.
(19) Lipeng Wan (College of Arts & Sciences): Optimizing ratio and performance for the compression + query use-cases.
(20) Da Yan (University of Alabama at Birmingham): Improving compression quality for graph data structure.
(21) Dong Dai (University of North Carolina at Charlotte): Analyzing the correlation between compression errors and post-hoc analysis metrics.
(22) Xiaoyi Lu (University of California, Merced): Compression for NVMe and distributed resilience proposal (SR-APPFL).
(23) Kibaek Kim (Argonne National Lab): distributed resilience proposal (SR-APPFL); compression for federated learning.
(24) Zhaorui Zhang (Hong Kong Polytechnic University): Compression for federated learning (both theoretical analysis and experiments).
(25) Chandrachur Bhattacharya (Division of Energy Systems, Argonne): Improving compression for TAPS MultiPhysics Datasets.
(26) Grosset Pascal (Los Alamos National Laboratory): Improving compression quality for Oct-Tree used in HACC.
(27) Milan and Michela from NCSU and Qtensor teams: Developing/optimizing lossy compression for Qtensor package
(28) Yanfei Guo (MPICH): Developing lossy compressor for MPICH.
(29) Jon Calhoun and Robert Underwood: Analysis of lossy compression for machine/deep learning applications and improving compression quality.
(30) Junjing Deng (APS): Assessment of lossy compression quality for many different users from various research institutes.
(31) Jiannan Tian (IU): Optimizing GPU performance for cuSZ (more specifically: cuSZ-i accepted by SC24).
(32) Jinyang Liu (U. Houston): Estimating compression quality for error-bounded lossy compressors.
(33) Milan Shah and Michela Becchi (NCSU): Improving performance for Deep learning on AI testbeds using lossy compression.
(34) Cheng Wang (EVS, Argonne): Improving lossy compression quality for Katrina Hurricane datasets.

10. Argonne CORE values

Impact: My research work, software and libraries gained a new level of impact in this year. I have 39 papers published/accepted by prestigious conferences and journals (including SC, IPDPS, HPDC, ICS, VLDB, ICDE, ICLR, ICML, DSN, ICDCS, IEEE VIS, etc.). We keep maintaining many software packages/libraries, including SZ2, SZ3, SZx, cuSZ, cuSZp2, Z-checker, QCAT, Libpressio, SZp, ROIBIN-SZ, MDZ, ZCCL, TZ, etc., as well as developing new libraries including Eureka, FFCz, OPAL, hZCCL2, etc. Statistics show that the unique year-to-date downloads reached a total of 5000+ for key packages/libraries. My published work gained about 1500 new Google citations compared with the last-year report (7371 citations -> ~9171 citations up to Sept 10, 2026). Google citation link: https://scholar.google.com/citations?user=zh3foWUAAAAJ&hl=en

Safety: I rigorously submitted all the 39 accepted/published papers (with PAM versions) to the Panda system for clearance, and also processed the copyright forms for each of them carefully with our Argonne copyright authority division (Argonne library). For any of these papers, I always tried best to make sure the correctness and efficiency of the design, implementation and appropriateness of the writing in the paper.

Respect: I fully respect every student and colleagues during the collaborative research. For the new students who lack experience in doing research or writing papers, I am always working very closely in overcoming the technical issues during the research, which is very helpful to their rapid growth in research. During any discussion of a research issue, I always listen carefully to different opinions of my students, my colleagues and my supervisor. Since some students

need to take classes during weekdays, I often schedule weekly meetings with them in the evening time or even on weekends. As long as any student stopped by my office, I always spend a lot of time discussing research problems with him/her.

Integrity: I actively keep the commitments, and take responsibility for the actions and outcomes regarding our projects and developed software. (1) For any of my papers, I always make sure the correctness and efficiency of the design, implementation and appropriateness of the writing in the paper. For example, I worked closely with my students to obtain all the three reprehensibility badges (artifact available, artifact functional, and results reproduced) for each of our papers accepted by SC26. (2) For the projects, I keep an active research and development progress for each of the projects I am leading and participating, and all the quarterly and annual reports have been approved by the program managers. (3) Regarding the software, I am co-developing and co-maintaining many libraries (the list can be found in Section 4). I actively responded to the questions and comments submitted to the corresponding github repo or sent to my email to keep a high integrity of the software function and performance.

Teamwork: I am currently (co)leading 6 projects as (co)PI and actively participating in 3 other projects as senior-personnel: see Section 3.3: Projects and Technical Leads, for more details. I kept a very wide range of collaborations: including 34 existing collaborations maintained from 2025 and 10 new collaborations established in 2026. More details can be found in Section 9: Collaboration. These collaboration involves many institutes around different countries, including RIKEN (Japan), INRIA (FR), University of Warwick (UK), Barcelona supercomputing center (Spain), Indiana University (USA), University of California, Riverside (USA), University of California, Merced (USA), Clemson University (USA), University of Chicago (USA), Stevens Institute of Technology (USA), Florida State University (USA), University of Florida (USA), Texas State University (USA), University of Minnesota Twin Cities (USA), etc. I co-mentored 30+ students in total at Argonne in 2025-2026. More details can be found in Section 7: Student mentoring.

11. Goals and plan for next year

- **Completing Genesis Mission Phase I Projects and Preparation for Phase II Proposals.** I am co-leading two partner-led Phase I Genesis Mission projects. My highest priority for the coming year is to successfully complete the research and development planned for both projects. We have established weekly meetings for each project and coordinated activities across all project partners. Both projects are progressing on schedule and remain on track.
- **Submit more proposals which were rejected this year to other opportunities (such as NSF).** I plan to submit two proposals to the NSF Future CoRe program and one proposal to the NSF AI Datasets program (NSF 26-512: *Unlocking Dataset Value for AI-Enabled Scientific Discovery*). I also plan to submit a proposal to the NSF CSSI program within OAC, provided that the solicitation remains active. In addition, I plan to submit two Genesis Mission Phase II proposals building on our two funded Phase I projects.
- **Continuous research serving DOE scientific applications.**

  I will keep maintaining and improving the existing projects and software.

  1) For general lossy compression software maintenance, I will also keep improving the reliability and performance for different compressors such as SZ3 - a loosely-coupled compression model, linear-regression-based compression, SZx, SZp, and (cu)SZp.
  2) I will keep leading/coleading three projects including my ECRP. I will mentor at least one student to explore a smart, efficient error-bounded lossy compression framework – SDR.
  3) I will explore more research strategies regarding lossy compression, such as matrix-vector supported homomorphic compression with my students, improving compression for Cerebras by Wavelet and RANS algorithm, etc.
  4) I will continue leading the SR-APPFL project, coordinating with our collaborators and students to ensure that the project progresses according to the timeline established in the proposal.
  5) I will continue working closely with other team members on the NSF CSSI FZ project and co-mentoring several long-term students involved in the project.
  6) I will continue leading Thrust 3 of the new DOE ZF project, led by Franck Cappello, while also collaborating closely with team members on the project's other thrusts.
  7) I will maintain collaboration with Xiaodong Yu and Yanfei Guo on using lossy compression to improve communication performance for MPICH and AI-for-science surrogate applications (such as APS)
  8) I will keep exploring how to improve federated learning performance by new techniques such as error-bounded lossy compression (Major collaborators include Kibaek Kim from MCS, Xiaodong Yu from SIT)
  9) I will keep exploring how to leverage lossy compression techniques to improve AI in different use-cases such as reducing memory footprint and reducing training data size/overhead. (Collaborator: Sian Jin)
  10) I will explore AI for science research, such as using LSTM+GA in the autonomous vehicle (AV) driving test application (collaboration is being planned with Guanpeng Li).

- **More collaboration**. I will keep providing lossy compression support (improve SZ's compression quality and performance) to diverse applications and projects, including Illumine/APS, Climate/OceanBox, Flamespeed, CFD, Net5000, etc. I will also keep a close collaboration outside Argonne campus, including Clemson U., IU, UCR, U. Chicago, U. of Iowa, UMN, INRIA, etc.
- **Better software**: Keep maintaining the multiple libraries (FSZ, IPComp, ZCCL, SZ, SZx, SZp/cuSZp, cuSZ, InterpSZ, MDZ, Z-checker, QoZ, FAZ, cuZ-Checker, z-checker-installer, QCAT, etc.) carefully. Evaluating our developed software/libraries on A21 systems (such as Aurora if available) or their prototypes.
- **Student mentoring:** I will continue actively mentoring students and early-career researchers through regular individual and project meetings. I will guide them in identifying research problems, designing and implementing solutions, conducting rigorous performance evaluations, developing software, and preparing high-quality publications. I will also support their professional development and encourage greater research independence and leadership.